PFC Boost Inductor — Production Engine Equation Reference

Step-by-step equations for every quantity computed by our production backend **step7_magnetic_calc.py** (Phase-1 v10). Transcribed from the code as of 2026-06-10. Counterpart record to the Simulation-Agent's PFC_Inductor_Engine_Equations.pdf, kept for side-by-side comparison before any merge.

Conventions. Distributed-gap powder toroid (ferrite path noted where it differs), CCM boost PFC, n_{ph}-phase interleaved. Geometry in mm; flux density in T; bias field H in A/m internally and oersted (Oe) for retention; Steinmetz frequency in Hz into the DB; resistance in Ω; power in W; temperature in °C. θ □ (0, π) is the line half-cycle angle; the engine integrates θ-dependent quantities on an **M = 360-point** midpoint grid. Subscript "single" = one core of the stack; stack totals = single × n_{stk}. **Material data (core loss $P_v(B,f,T)$ and DC-bias k(H)) comes from the MagneticsDB as temperature-aware interpolated curves**, not closed-form fits — this is our key accuracy lever and is shown below as DB(·).

S0 — Physical & model constants

(E1) $\mu_0 = 4\pi\times10^{-7}$ **H/m,** $\rho_{Cu,20} = 1.72\times10^{-8}$ **Ω·m,** $\alpha_{Cu} = 0.00393\ K^{-1}$

- copper resistivity at 20 °C and its temperature coefficient
- $\rho_{Cu}(T) = \rho_{Cu,20}(1 + \alpha_{Cu}(T - 20))$

(E2) 1 Oe = 79.577 A/m

- oersted conversion used wherever the DB bias curve is addressed in Oe

(E3) iGSE: $c = 1.444$, $b = 2.106$, $l_c = 3.5435$, $K_{iGSE} = 2^c / ((2\pi)^{c-1}\cdot l_c)$

- fixed improved-Generalised-Steinmetz constants for the triangular-ripple duty correction F(D)

(E4) Proximity (Dowell-like, v10): $k_{skin} = 0.50$, $k_{prox} = 0.40$, $k_{crowd} = 0.25$

- calibrated coefficients for the litz/TIW AC-resistance model (E20)

(E5) Thermal 2-node split (v10): $s_C = 1.00$, $s_W = 0.90$, **couple = 0.50, hotspot = 1.12;** $l_{lead} = 150$ **mm**

- core/winding ambient-path split, core–winding coupling, interior hotspot factor; default lead length added to copper length

S1 — Unbiased inductance, required retention, and turns (powder)

(E6) $L_0 = A_{L,single} \cdot n_{stk} \cdot N^2 \times 10^{-9}$ **[H]**

- $A_{L,single}$ — inductance factor of ONE core at zero bias [nH/T^2]; N — turns

(E7) $L_{0,min} = L_0(1 - tol_{AL})$, $L_{0,max} = L_0(1 + tol_{AL})$ $(tol_{AL} \approx 0.08)$

(E8) $k_{req} = L_{target} / L_0$ **(required bias retention at the worst DC point)**

- turns are chosen as the smallest N whose biased inductance still meets the target:
- find N: DB $k(H_{worst}(N)) \cdot L_0(N) \geq L_{target}$

(E9) $I_{dc,worst} = \max_{OPS} [\ \sqrt{2}\cdot P_{out} / (\eta\cdot V_{in}\cdot PF\cdot n_{ph})\ ]$

- worst-case per-phase crest current scanned over all 9 operating points (OPS)

Note: Ferrite path instead converges turns + gap l_g to a target peak flux (E14 uses the gapped reluctance).

S2 — DC-bias field and DB permeability retention

(E10) $H_{dc} = N \cdot I / l_{e,single}$ **[A/m],** $H_{Oe} = H_{dc} / 79.577$

- uses l_e of ONE core (NOT $n_{stk}\cdot l_e$) for stacked toroids

(E11) k(H) = DB.get_k_bias(material, H_{Oe}) [–]

- material-specific DC-bias curve interpolated from the MagneticsDB (per material, per H)

Note: Fast inner waveform loop uses an EDGE-family closed-form fallback $k(H) = \min(1, [1/(0.01 + 9.202\times10^{-10}\ H^{3.044})]/100)$.
KNOWN LIMITATION: this fallback is EDGE-calibrated; non-EDGE materials should route through DB.get_k_bias.

(E12) $L(H) = k(H) \cdot L_0$ **(biased inductance)**

S3 — Operating-point currents (per OPS corner {V_{in}, P_{out}, η, PF})

(E13) $P_{in} = P_{out} / \eta$, $V_{pk} = \sqrt{2}\cdot V_{in}$, $D_{pk} = 1 - V_{pk} / V_{out}$

- V_{out} = 393 V boost rail; D_{pk} — duty at the line crest

(E14) $I_{\varphi,avg,crest} = \sqrt{2}\cdot P_{out} / (\eta\cdot V_{in}\cdot PF\cdot n_{ph})$

- per-phase switching-average current at the line crest

S4 — Line-cycle waveforms over θ □ (0, π), M = 360 midpoints

(E15) $\theta_n = (n + ½)\pi/M$, $v_{in}(\theta) = V_{pk} \sin\theta$, $D(\theta) = \mathrm{clip}(1 - v_{in}/V_{out}, 0, 0.98)$

(E16) $i_{avg}(\theta) = I_{\varphi,crest} \sin\theta$, $H(\theta) = 0.4\pi\cdot N\cdot i_{avg}(\theta) / l_{e,cm}$, $L(\theta) = k(H(\theta))\cdot L_0$

(E17) $\Delta I_{pp}(\theta) = v_{in}(\theta)\cdot D(\theta) / (L(\theta)\cdot f_{sw})$, $I_{hf}(\theta) = \Delta I_{pp}(\theta) / (2\sqrt{3})$

- triangular switching ripple peak-to-peak and its RMS

(E18) $B_{ac,pk}(\theta) = v_{in}(\theta)\cdot D(\theta) / (2\cdot N\cdot A_e\cdot f_{sw})$, $B_{dc}(\theta) = L(\theta)\cdot i_{avg}(\theta) / (N\cdot A_e)$, $B_{max}(\theta) = B_{dc} + B_{ac,pk}$

- $A_e = A_{e,single}\cdot n_{stk}\times 10^{-6}$ [m²]

S5 — Flux density, radial crowding, saturation margin

(E19) $B_{ac,pk} = V_{pk,90}\cdot D_{pk,90} / (2N\cdot A_e\cdot f_{sw})$, $B_{dc} = L_{target}\cdot I_{\varphi,crest} / (N\cdot A_e)$

- headline flux evaluated at the 90 Vac low-line reference corner

(E20) $B_{max,FL} = B_{dc} + B_{ac,pk}$, $B_{min,FL} = B_{dc} - B_{ac,pk}$

(E21) $crowd_{ax} = r_{mean} / r_{in} = [(ID/2 + OD/2)/2] / (ID/2)$, $B_{inner} = B_{max,FL} \cdot crowd_{ax}$

- toroid 1/r inner-bore flux concentration (v10)

(E22) $margin_{sat} = (B_{sat}(T_{core}) - B_{max}) / B_{sat} \times 100\ \%$

- $B_{sat}(T_{core})$ from the DB at the converged core temperature; inner-bore margin checked too

(E23) Ferrite: $B_{dc} = \mu_0 N\cdot I / (l_g + l_e/\mu_r)$

- gapped-reluctance bias flux; μ_r = DB.get_mu_r(material, 100°C); l_g — air gap

S6 — Copper geometry, MLT, and DC resistance

(E24) $w_{core} = (OD - ID)/2$, $MLT = 2(w_{core} + HT\cdot n_{stk}) + 2\cdot OD_{bundle}$ [mm] (v10)

- v10 uses the actual bundle OD as the build allowance (legacy form used a fixed 3.8 mm)

(E25) $l_{Cu} = N\cdot MLT/1000 + l_{lead}/1000$ [m]

- lead wire (entry + exit, 150 mm) added to the wound length

(E26) $R_{DC}(T) = R'_{20}\cdot(1 + \alpha_{Cu}(T - 20))\cdot l_{Cu}$ [Ω]

- R'_{20} — catalog wire resistance per metre at 20 °C [Ω/m]

(E27) $FF_{Cu} = N\cdot A_{Cu} / W_{a,bore}$

- bare-copper bore-window fill; $W_{a,bore}$ is the single-core window area (invariant with stacks)

S7 — Skin depth and AC resistance (physics-based Dowell-proximity, v10)

(E28) $\delta = \sqrt{(\rho_{Cu}(T) / (\pi\cdot f_{sw}\cdot\mu_0))}$, $x = d_{strand} / (2\delta)$

- δ — skin depth at f_{sw}; x — strand-radius / skin-depth ratio

(E29) $F_{skin} = 1 + k_{skin}\cdot x^2$, $F_{prox} = 1 + k_{prox}\cdot(N_{lay}-1)\cdot x^2\cdot(1 + k_{crowd}(crowd_{ax}-1))$

- inter-layer proximity grows with layer count and inner-bore crowding

(E30) $R_{AC}/R_{DC} = \max(1, F_{skin}\cdot F_{prox})$

- litz/TIW path; solid/enamel uses an exact Bessel skin factor (compute_dowell_factor)

Note: This physics-based ratio is a strength of our engine vs. a flat 1.15 heuristic.

S8 — Bore layer count and residual window

(E31) OD_{bundle} — catalog bundle OD (authoritative) or computed from strands & fill

- feeds both MLT (E15) and the layer/proximity model (E22)

(E32) N_{lay}, t_{pl}, r_{bore} = layers($N \cdot n_{par}$, ID, OD_{bundle}); $hole_{ID} = \max(0.5, 2 \cdot r_{bore})$

- turns-per-bore-layer and residual bore radius after all layers (negative overfull)

S9 — Core loss (DB Steinmetz, iGSE duty correction, cycle-averaged)

(E33) $P_v(B, f, T)$ = DB.get_core_loss(material, f_{sw}, $B_{ac,pk}$, T) $[W/m^3]$

- temperature-aware loss density interpolated from the MagneticsDB material curves

(E34) $F_D(D) = K_{iGSE} \cdot (D^{1-c} + (1-D)^{1-c})$, D clipped to [0.02, 0.98]

- iGSE duty-ratio correction for the asymmetric triangular flux

(E35) $p_{core}(\theta) = P_v(B_{ac,pk}(\theta), f_{sw}, T_{core}) \cdot F_D(D(\theta)) \cdot V_e$, $P_{core} = (1/M)\ \Sigma_n\ p_{core}(\theta_n)$

- $V_e = V_{e,single} \cdot n_{stk}$ $[m^3]$; this cycle-averaged value is the authoritative P_{core}

(E36) Uncertainty band: $P_{unc,lo} = P_{Cu,100} + 1.05 \cdot P_{core}$, $P_{unc,hi} = P_{Cu,100} + 1.20 \cdot P_{core}$

- +5 % to +20 % core-loss uplift band (unanchored until FEA/bench calibration)

S10 — RMS decomposition and copper loss (cycle-averaged)

(E37) $I_{\varphi,rms} = \sqrt{(1/M)\Sigma\ i_{avg}(\theta)^2}$, $I_{hf,rms} = \sqrt{(1/M)\Sigma\ I_{hf}(\theta)^2}$

(E38) $P_{Cu}(T) = I_{\varphi,rms}^{\ 2} \cdot R_{DC}(T) + I_{hf,rms}^{\ 2} \cdot R_{AC}(T)$

- $R_{AC}(T) = R_{DC}(T) \cdot (R_{AC}/R_{DC})$; reported at T_{core}, 25 °C and 100 °C

(E39) $J = I_{\varphi,rms} / (A_{Cu} / n_{par})$ $[A/mm^2]$

- copper current density per parallel conductor

S11 — Thermal (SA single-node convergence + v10 two-node hotspot)

(E40) $SA = [\ \pi \cdot OD_w \cdot OH + (\pi/2)(OD_w^{\ 2} - hole^2) + \pi \cdot hole \cdot OH\ ] / 100$ $[cm^2]$

- wound-part exposed surface area; OD_w, OH from the assembled wound envelope

(E41) $\Delta T_{SA} = (P_{tot} \cdot 1000 / SA)^{0.833}$ [K]

- natural-convection surface-area power law

(E42) Convergence: iterate $T_{core} \leftarrow T_{amb} + \Delta T_{SA}(P_{core}(T_{core}) + P_{Cu}(T_{core}) + P_{fring})$ until $|\Delta| < 0.2$ K

- couples temperature-dependent core & copper loss to the surface model (10-iteration loop)

(E43) $\theta = \Delta T_{SA}/P_{tot}$; $R_{ca} = \theta(s_C+s_W)/s_C$, $R_{wa} = \theta(s_C+s_W)/s_W$, $R_{cw} = \theta \cdot couple$

- two-node resistances derived from the calibrated SA baseline

(E44) Solve KCL 2×2 for ΔT_c, ΔT_w; $\Delta T_{hotspot} = \max(\Delta T_c, \Delta T_w) \cdot 1.12$

- core/winding node rises and interior hotspot (1.12 over node average)

S12 — Gap fringing (ferrite only)

(E45) P_{fring} = Rogowski(I_g, A_e, W_a, N, d_{bundle}, $R_{DC,100}$)

- gap-fringing eddy loss in winding turns near the gap (compute_rogowski_fringing); powder cores: 0

S13 — Per-V_{in} screening table (analytical, single-point)

A faster single-point analytical pass over all 9 OPS corners drives the per-V_{in} sweep table and charts. It is intentionally lighter than the S4/S9/S10 waveform integration:

(E46) $B_{ac,pk} = V_{pk}D_{pk}/(2N \cdot A_e f_{sw})$, $P_{core} = P_v(B_{ac,pk},f,T) \cdot F_D(D_{pk}) \cdot V_e$, $P_{Cu} = I_{rms}^{\ 2}R_{DC}(T) + I_{hf}^{\ 2}R_{AC}(T)$

- single crest-point evaluation per corner (no θ integration)

Note: IMPORTANT: this table is a SCREENING estimate and differs from the authoritative S4/S9/S10 waveform totals at the same V_{in}. The Review page anchors this table to the waveform result at 90 V / 100 °C so the two presentations agree (P_{core}, P_{tot} ratios).

S14 — Worst-case scan, acceptance, and ranking score

Every OPS corner is evaluated; the design reports worst-case total loss, B_{max} (mean path) and B_{inner}, minimum guaranteed L, maximum ΔT and J. Pass/fail uses $L \geq L_{target}$, B_{max} and inner-bore vs $B_{sat}(T)$, $FF_{Cu} \leq$ limit, $\Delta T \leq$ budget, J in band.

(E47) $$\text{score} = w_{loss} \cdot P_{tot}/P_{ref} + w_{vol} \cdot V_e/V_{ref} + w_{\Delta T} \cdot \Delta T/\Delta T_{ref} + w_{cost} \cdot (\ldots) + w_{fill} \cdot (\ldots)$$

- composite ranking score (lower is better); candidates returned top-N per stack count

Appendix — Variable glossary (code names)

Symbol	Code name (step7_magnetic_calc.py)	Unit	Meaning
mu0	MU0	H/m	Permeability of free space
rhoCu,alphaCu	RHO_CU_20 / ALPHA_CU	ohm*m, 1/K	Copper resistivity / temp coefficient
c,b,Ic,K	IGSE_C / IGSE_B / IGSE_IC / IGSE_K	-	iGSE duty-correction constants
kskin,kprox,kcrowd	_PROX_kSkin/kProx/kCrowd	-	Dowell-proximity coefficients (v10)
sC,sW,couple,hotspot	_THERM_sC/sW/couple/hotspot	-	Two-node thermal split + hotspot factor
N	N	turns	Number of turns (chosen by _turns_powder/_turns_ferrite)
nstk	stacks	-	Stacked core count
AL,single	AL_nom_nH	nH/T2	Inductance factor of ONE core
L0(min/nom/max)	L0_min/nom/max_uH	uH	Unbiased inductance + AL tolerance band
kreq	kreq_nom	-	Required retention L_target/L0
Ae,single/Ve,single	Ae_single_mm2 / Ve_total_cm3	mm2/cm3	Single-core area / total volume
Le,single	Le_single_mm	mm	Magnetic path length of ONE core
Wa,bore	Wa_single_mm2	mm2	Single-core bore window area
Bsat(T)	Bsat_at_Tcore	T	Saturation flux at core temperature (DB)
k(H)	DB.get_k_bias / _retention_edge	-	DC-bias retention (DB curve; EDGE fallback)
Pv(B,f,T)	DB.get_core_loss	W/m3	Temperature-aware core-loss density (DB)
Hdc/HOe	H_Oe_design / H_Oe_worst	A/m, Oe	DC bias field
Iphi,crest	I_phi_avg_crest	A	Per-phase crest average current
Dpk	Dpk90	-	Duty at line crest
dIpp/Ihf	dIL_pp_A / IhfRms_A	A	Switching ripple pk-pk / RMS
Bac,pk/Bdc/Bmax	Bac_pk_T / Bdc_T / Bmax_FL_T	T	AC peak / DC / mean-path peak flux
crowd_ax/Binner	crowd_axial / Bmax_inner_FL_T	-, T	Inner-bore crowding / crowded peak flux
MLT	MLT_v10_mm	mm	Mean length per turn (2(w+HT*stk)+2*ODb)
lCu	Cu_length_m	m	Total conductor length incl. leads
RDC(T)	DCR_25C/100C_mOhm	mohm	DC resistance at temperature
RAC/RDC	Rac_Rdc	-	AC/DC ratio (Dowell-proximity or Bessel)
delta/x	(internal)	m, -	Skin depth / strand-to-skin ratio
Nlay,tpl,rbore	layers_needed / (internal)	-	Bore layers, turns/layer, residual radius
FFcu/Ku	FFcu / Ku	-	Window fill / bore utilization
FD(D)	_Fd(D)	-	iGSE duty correction
Pcore	Pcore_W	W	Cycle-averaged core loss (authoritative)
Punc lo/hi	P_unc_lo_W / P_unc_hi_W	W	Core-loss uncertainty band
Iphi,rms/Pcu	Ihf_rms_A / Pcu_100C_W	A, W	RMS current / copper loss
J	J_A_mm2	A/mm2	Copper current density
SA/dT	SA (internal) / dT_rise_C	cm2, K	Wound surface area / surface rise
dTc/dTw/dThot	dT_core_C/dT_wdg_C/dT_hotspot_C	K	Two-node rises + hotspot
Rca/Rwa/Rcw	(returned by _two_node_thermal)	K/W	Node thermal resistances
Pfring	P_fringing_W	W	Gap fringing loss (ferrite)
Ptotal(25/100)	Ptotal_25C_W / Ptotal_100C_W	W	Total loss at temperature
score	score	-	Composite ranking score (lower better)

Honest status note. Strengths vs. the Simulation-Agent reference engine: temperature-aware DB core-loss and DC-bias curves (E10, E25), a physics-based Dowell-proximity R_{AC} (E23) rather than a flat 1.15, a two-node hotspot thermal model (E48–E49), *and ferrite gap fringing (E50). Open items: the inner waveform loop's k(H) fallback is EDGE-calibrated (route non-EDGE through DB.get_k_bias); the per-V_{in} screening table (S13) is single-point and is anchored to the waveform result on the Review page; R_{AC}, thermal and bias retention are still first-order until FEA/bench data is attached. All other quantities are exact arithmetic on the DB + designer-selected geometry/wire.*